SPEA

Electronic Board Testing

3030CE

Multi-Core Multi-Function Bed of Nails Tester



Product Description

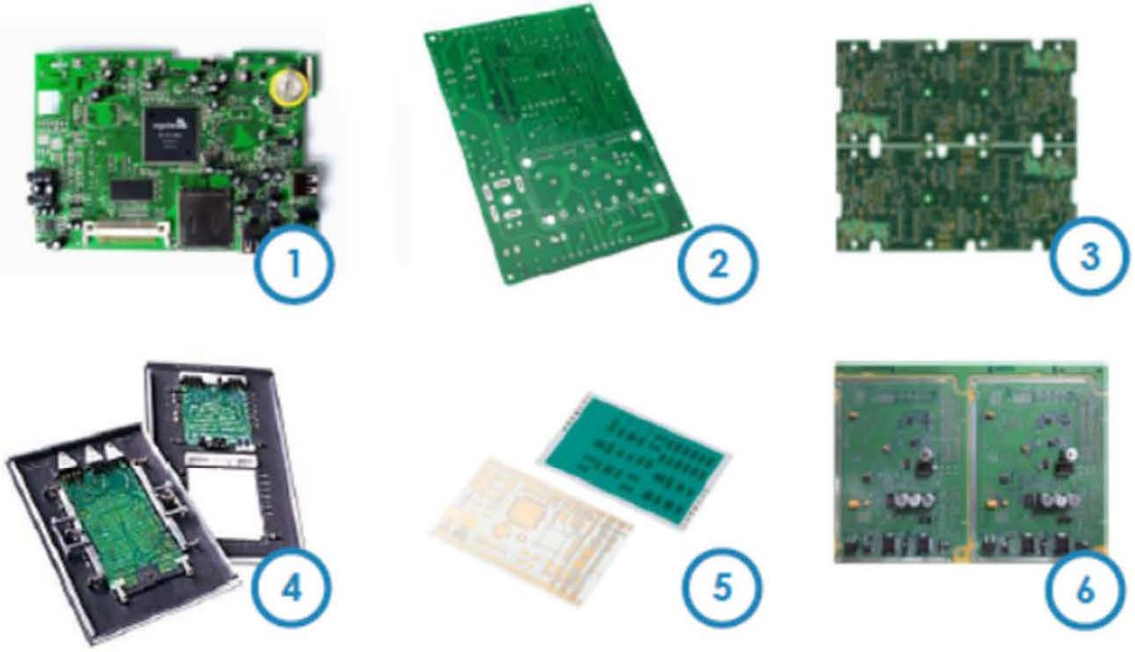
Version: V1 – 27.10.2020
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Main Features:

- **Multi-Core Architecture**: up to 2x greater throughput and up to 2x board parallel testing, also with different Test Programs.
- **Full Test Capabilities** :
  - In-Circuit Test Power OFF and Power ON
  - Short Circuit Test and Open Pin Scan
  - V/I Spike Detector
  - Net Insulation, Impedance and Voltage Test
  - Light Tests (Chromaticity and Light Intensity)
  - Boundary Scan Test (XJTAG, JTAG Technologies Symphony, Goepel)
  - Multi-Device Parallel On Board Flashing
  - Power On Test
  - Digital and FunctionalTest
  - High Power Functional Test
  - Built-In Self-Test (BIST)
- **Wide range of applications**: PCBA and PCB, Panels of boards, UUTs on pallet or Test Jig.
- **Automatic Test Program Generation**: Leonardo YA ICT Operating System allows to **quickly and easily develop the Test Program** even to unexperienced users.
- **Compatibility with the entire 3030 family**: easy migration of the test program and spare parts compatibility.
- **Very compact footprint** : 1.23m².

Application

1. Printed Circuit Board Assembled (PCBA)
2. Printed Circuit Board (PCB)
3. Panel of Boards
4. UUT on Pallet or Jig
5. Ceramic Circuit Board
6. Sticky Board



Key Features

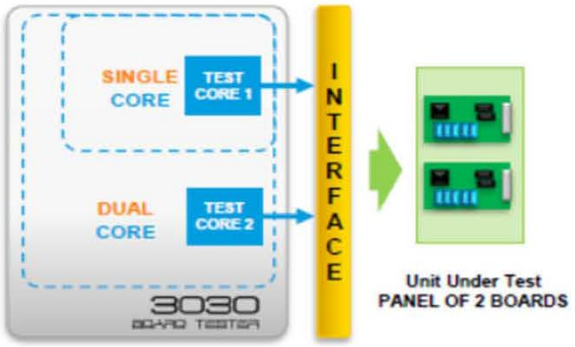
Multi-Core Architecture	• Up to 2x independent test core with dedicated CPU, instrumentation and memory • 2x independent Test Cores allows simultaneous execution of tests on 2 UUTs
Multi-Function Architecture	• Multiple test techniques on a single system (ICT, Optical, OBP, BIST, etc…)
Parallel On Board Device Flashing	• OBP module with 4x 8-channel flashing core • Up to 32x OBP modules installed in one tester for a total of 2048x channels dedicated to device flashing • Simultaneous Multi-site and Simultaneous Multi-die flashing • Dedicated CPU for I/O command and memory for programming data (no data transfer required)
True Per-Pin Architecture	• Each system channel is configurable by test program, and every nail can be used to perform any kind of test.
Cable-less Instrument Connection	• Direct connection between instruments and system interface. • Signal integrity is guaranteed without any leakage and crosstalk.
PC-Independent Architecture	• Test Program run from a pre-loaded System Core. • Test speed is determined by the System Core, not the System PC. • PC change/update does not require re-debug operations.
Fast Test Program Generation	• Debug & tuning of the test program is fully automatic and fast. • No need for test engineer or expert technician to develop and debug a test program.
Fully Upgradeable & Customizable	• Possibility to equip the tester with external cabinet to increase test capabilities and productivity with instrumentation, such as AC/DC generators, Active Loads, Power Matrix, third party instruments etc.

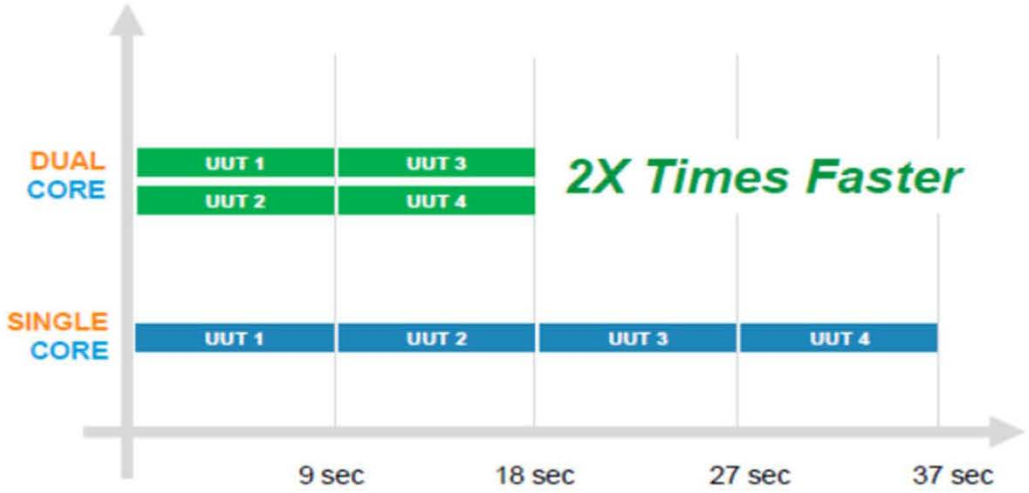
Multi-Core Architecture

The 3030CE testers can be equipped with **up to 2x Test Cores** that operate in parallel to **simultaneously test 2 boards/panels**.

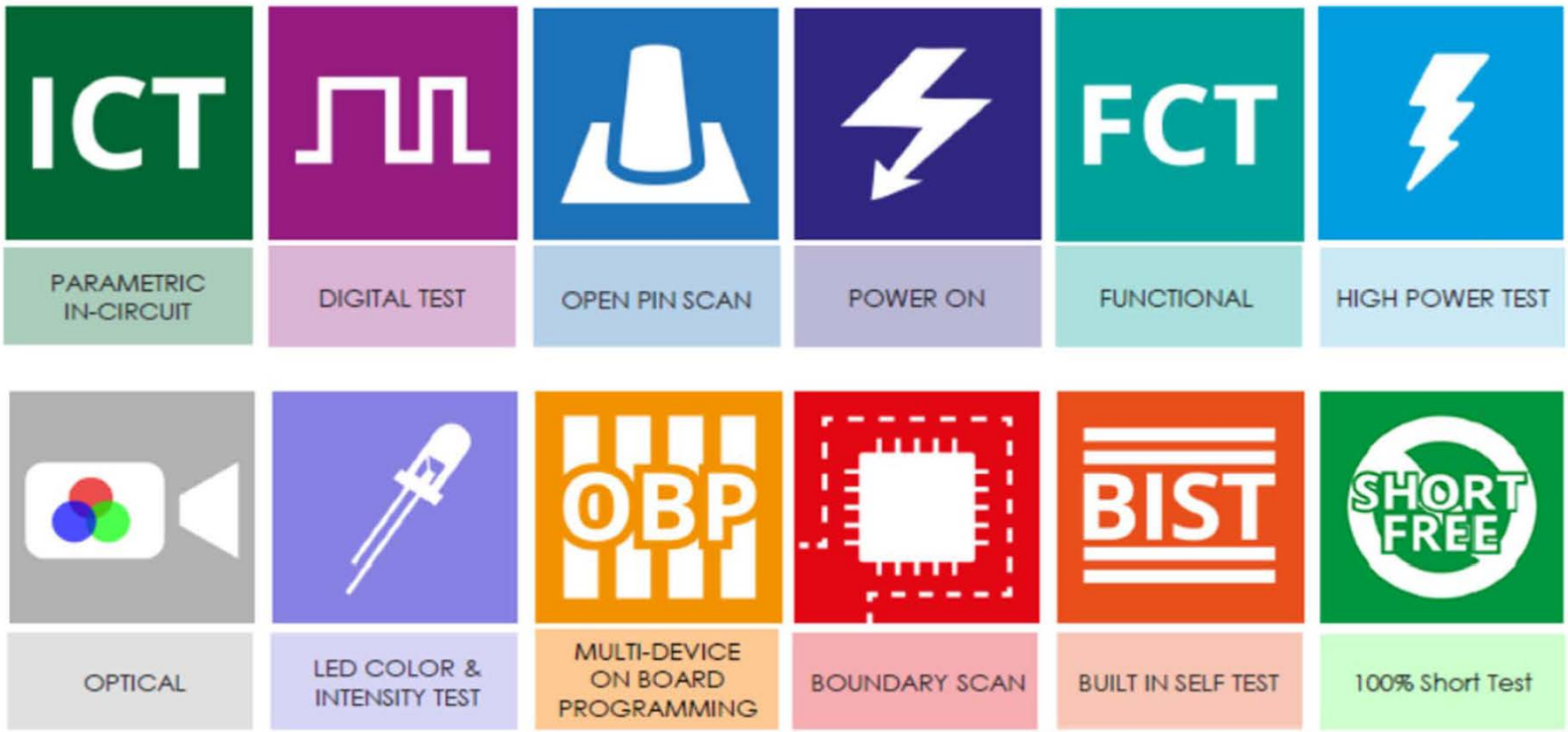
Each core includes **independent CPU**, **local memory** and **instrumentation**.

The instrumentation in each Test Core is related to the type of test that should be performed.



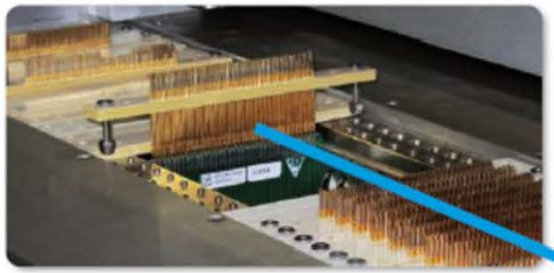


Multi-function Architecture

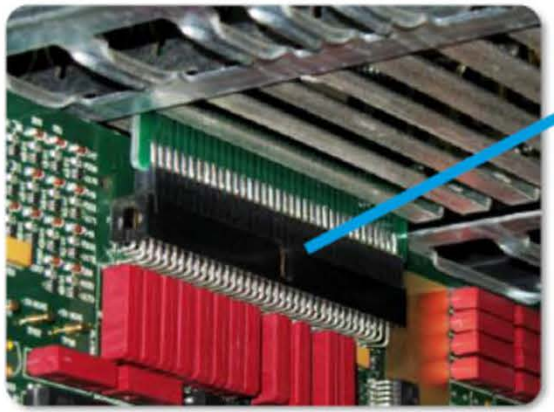


Cableless Instrument Connection

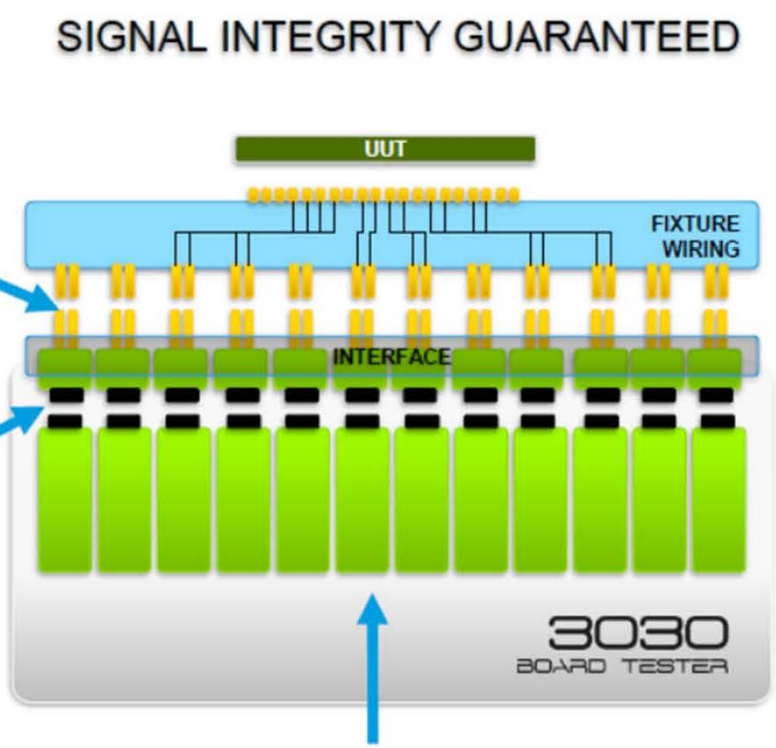
The 3030CE is conceived to provide direct connection between instruments and system interface pins. In this way signal integrity is guaranteed without any leakage and crosstalk.



System Interface Pins



Instrument Output Connector

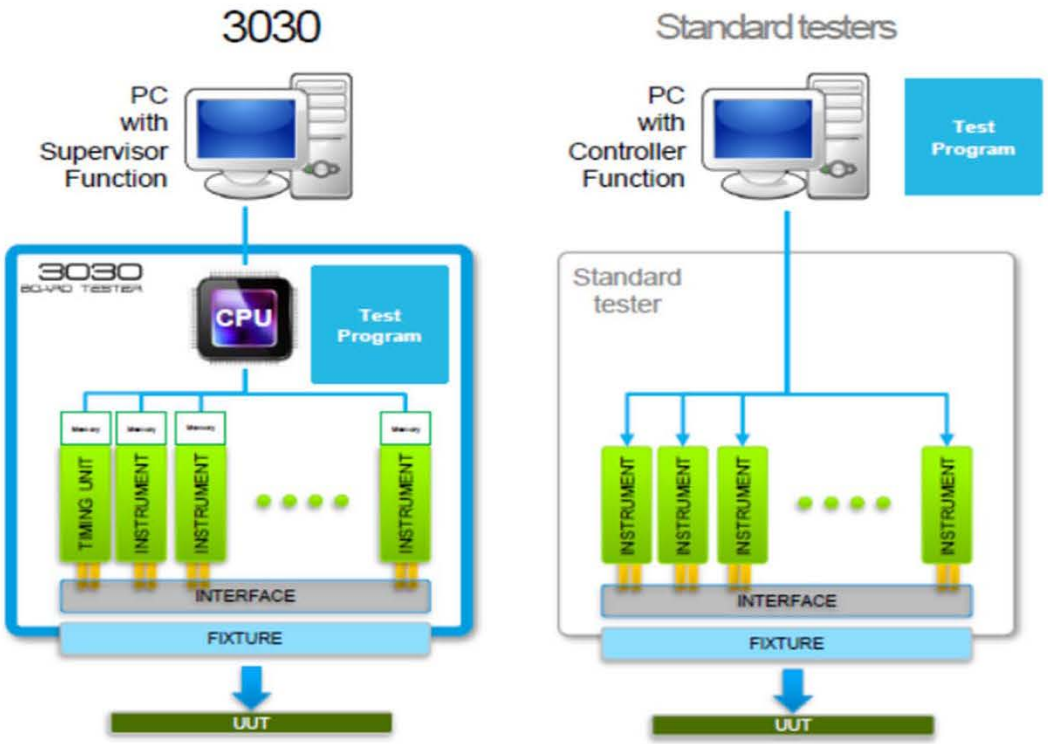


Instrumentation

PC-Independent Architecture

The SPEA **Instrumentation Controller Architecture** is conceived to pre-load and run the Test Program from Test Core CPU. With this architecture the **test speed is determined by the Test Core CPU**, which is independent from System Controller.

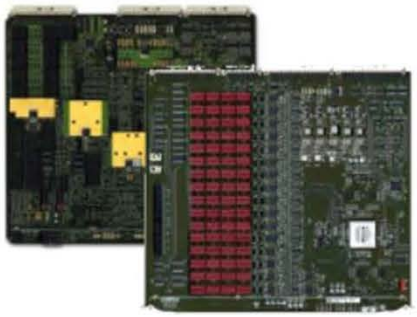
Competitor's testers control the instrumentation with the PC. With this architecture the test execution speed is determined by the PC and other running applications (e.g. antivirus). Furthermore, if the PC needs to be changed, re-debug of all applications will be required.



Full Upgradeability & Customizable

Each system can be **factory equipped** or **upgraded on field** with all instrumentation needed to reach full coverage, such as:

- Power instrumentation (as AC/DC generators, Active Loads, Power Matrix, etc.).
- Third party instruments to increase test capabilities and productivity.





16-bit instrumentation

Power instrumentation

Third Party HW



Product Composition



ID	Item
1.	Receiver
2.	Instrumentation Rack *(not visible)*
3.	System Controller *(not visible)*
4.	Leonardo ICT YA Operating System
5.	Extension Cabinet *(option)*

Receiver

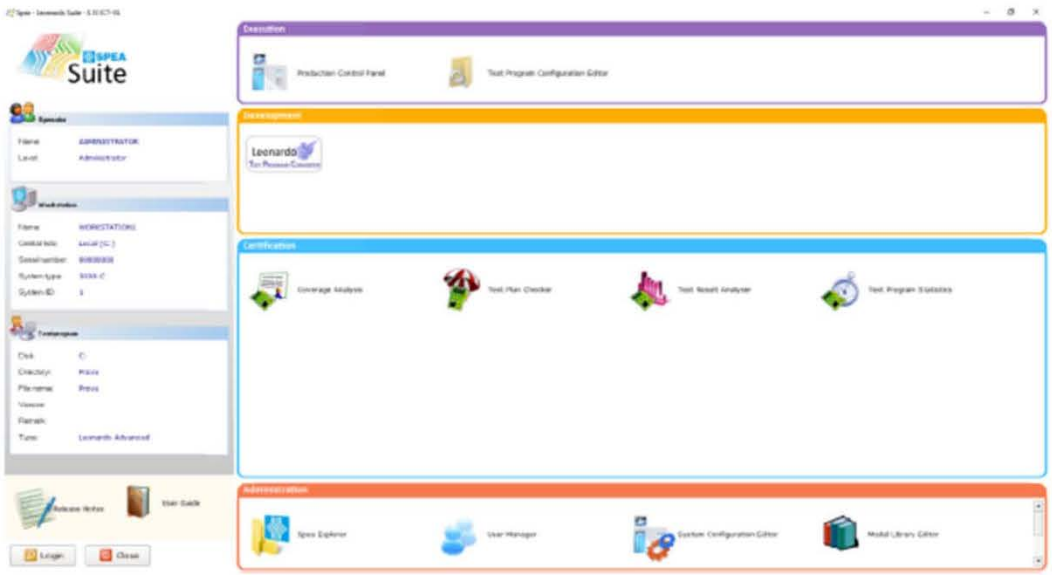
The 3030CE can equipped with Drawer/Motorized Actuation Receivers compatible with Fixture designed for 3030IL Tester and 3030X (single interface) Testers.

SPEA Receivers - Tech Specs*	
Actuation Type	**Drawer/Motorized**
Interface Test Points no.	Up to 2048
Interface no.	Up to 2
Inteface Type	ZIF
Test Area (LxW) [mm]	540x380
Pressure force [kg]	300
Pressing levels	2 (Fixed)
Test area protection	Cover/Door/ Optical Barrier

(*) Refer to *3030 Receiver - Product Article Overview* for Receiver specifications.



Leonardo ICT YA



Key Features

❑ **Simple and intuitive**

- Ease of use, quick and intuitive also for not skilled users.
- Very short learning curve.
- No need for additional documentation or training.
- All the tests are easy to be developed and managed by any technician even if not skilled.

❑ **Very Fast Test Program development and debug**

- Up to 95% of the Test Program can be automatically generated.
- Test specification approach.
- Consistent time saved using tools to automate repetitive operations.

Test Type Overview

Discharge Capacitor Test	• Discharge Capacitor Test to avoid damages of the machine and the UUT due to charged capacitors.	
Short Circuit Test	• Short Circuit Test – All Nets • Short Circuit Test – Low Impedance	
ICT Power Off	• Safe ICT Power Off - @max 0.3V-5mA • Standard ICT Power Off - @max 10V-100mA • Hi-Accuracy ICT Power Off @ 0,1% Accuracy	• Hi-V ICT Power Off - @ 10V-100V • Stress ICT Power Off - @ up to 3A • Dynamic V/I ICT Power Off - @ 1µs V/I Timing
Open Pin Test	• Open Pin Test - Junction Scan • Open Pin Test - Electro Scan	
Power Supply Test	• Supply Current Test • Internal Voltage Test • Voltage Regulators Test	
ICT Power On	• Standard ICT Test • Analog IC Test • Digital IC Test • Oscillator ICT Test	
Node Voltage Test	• Node Voltage Test	
Optical Test	• Presence Test - Parts not ICT Tested • Presence Test • XY Alignment Test • Character Recognition Test	
Light Test	• LED Test • Display Test	
On Board Programming	• On Board Programming	
Boundary Scan Test	• Boundary Scan Test	
Functional Test	• Functional Test	

Short Circuit Test Overview

All Nets Shorts Test	• Detect shorts between all nets.
Adjacent Pins Shorts Test	• Detects shorts between nets characterized by low impedance (below 1 ohm)

ICT Power Off Test Overview

Safe ICT Power Off @max 0.3V-5mA	• Safe In-Circuit Test. • To be used to test circuits using IC technology that do not allow V/I application higher than 5mA and 0,3V.
Standard ICT Power Off @max 10V-100mA	• Standard In-Circuit Test. • V/I maximum values used are 100mA and 10V.
Hi-Accuracy ICT Power Off @ 0,1% Accuracy	• High-Accuracy In-Circuit Test. • Measurements made using instrumentation with accuracy of 0,1%.
Hi-V ICT Power Off @ 10V-100V	• High-Voltage In-Circuit Test. • 4-Quadrant V/I force and measure up to 100V and 100mA.
Stress ICT Power Off @ up to 1A	• High-Current Stress In-Circuit Test. • 4-Quadrant V/I force and measure up to 3A. • Used to detect defects of power devices such as MOS and Power ICs.
Dynamic V/I ICT Power Off @ 1uS V/I Timing	• Dynamic In-Circuit Test. Safe Energy Controlled In-Circuit Test. • Force and measure synchronized V/I signals with timing of 1us step. • Used to test dynamically transformers, relays, power devices.

Open Pin Test Overview

Junction Scan Test	• Test of open pins of Digital ICs. • Measurement done testing the clamp diode present on most of Digital IC pins.
Electroscan Test	• Test of open pins of digital ICs, analog ICs and connectors. • Measurement done using Electro-Scan technique.

Power Supply Test Overview

Supply Current Test	• Automatic test of the UUT current sink at power-on.
Internal Voltage Test	• Parametric and functional test of the voltage regulators mounted on board.
Voltage Regulators Test	• Parametric Voltage Regulators Test with current sink @ operating or limit conditions.

ICT Power On Test Overview

Standard ICT Test	• In-Circuit Test of comparators and OPAMPs.
Analog IC Test	• In-Circuit Test of analog and mixed-signal ICs.
Digital IC Test	• In-Circuit Test of digital ICs.
Oscillator ICT Test	• Test of Oscillator up to 100MHz

NVT Test Overview

Node Voltage Test	• Node Voltage Test. • Test of the voltage, or measurable signals, of each net. • The test is done at the Reset status and, additionally, after any command send.

Optical Test Overview

Presence Test Parts not ICT Tested	• Optical Test. • Presence test only of the components not tested with ICT.
Presence Test	• Optical Test. • Presence test of the components.
XY Alignment test	• Optical Test. • XY alignment test of the components.
Character Recognition Test	• Optical Test. • Test of characters string.

Light Test Overview

LED Test	• Light test of the LEDs.
Display Test	• Light test of the Displays.

On Board Programming Overview

On Board Programming	• On board programming of programmable devices and flash uploading. • Independent 4x core included on each OBP module, each one with dedicated channels and memory. • Parallel programming of identical or different devices (up to 256).

Boundary Scan Test Overview

Boundary Scan Test	• Boundary Scan Test using JTAG, GOEPEL or XJTAG

Functional Test Overview

Functional Test	• Communication Port Pattern generator (CAN, RS232, USB, I²C-Bus, etc…) • Board electrical characteristics and component functioning.

Dimensions



Length:	1270 mm
Width:	670 mm
Height:	1560 mm
Footprint:	0.85 m²
Weight:	about 320 kg (Zif interface version)

Tester Characteristics

Test Area*	• Up to 540 x 380 mm (21" x 16.1")
SPEA Operating System	• Leonardo YA ICT
System Controller Unit	• Industrial PC Intel Core i7, Ram 8GB, n.2 HD Solid State
PC Operating System	• Windows 64 bits
Test Interface	• Up to 2048 channels

(*) Depending on the Receiver in use with the tester

Electrical and operating requirements

Item	Value
Mains	Single Phase 120÷240Vac ±10% Three Phase 380Vac ±10%
Mains power	2kVA typ. 15kVA max.
Frequency	50 ÷ 60Hz ±1%
Operating temperature	15°C ÷ 32°C (59° ÷ 90°F)
Operating Humidity	≥20% ÷ ≤70%
Storage Humidity	≤90%

ICT – In-Circuit Test

What does ICT do?

ICT electrically accesses test points on the PCBA using a **bed-of-nails fixture** and measures individual components, nets and circuit conditions.

Typical tests:

- Resistance
- Capacitance
- Inductance
- Diode
- Continuity
- Open circuit
- Short circuit
- Voltage
- Current
- Component presence/value
- Transistor checks
- IC-related tests
- Boundary scan/JTAG where supported

1. Comparison

Parameter	ICT	Functional Test (FCT)	Flying Probe	AOI	X-Ray
Full form	In-Circuit Test	Functional Test	Flying Probe Test	Automated Optical Inspection	X-Ray Inspection
Main purpose	Check components & electrical connections	Verify product functionality	Electrical testing without fixed fixture	Visual inspection	Internal solder/component inspection
Contact with PCB	Yes	Usually yes	Yes, probes move	No	No
Dedicated fixture	**Yes**	Usually yes	**No dedicated bed-of-nails fixture**	No	No
Test speed	**Very high**	Medium	Low–medium	Very high	Medium
Component value test	**Yes**	Sometimes	**Yes**	No	No
Open/short test	**Yes**	Limited	**Yes**	Visual only	Some defects visible
Wrong component	**Yes**	Sometimes	**Yes**	Some cases	Sometimes
Solder joint inspection	Limited/electrical	Indirect	Limited/electrical	**Excellent for visible joints**	**Excellent internally**

BGA inspection	Limited	Indirect	Limited	Limited	**Excellent**
Circuit operation	Limited	**Excellent**	Limited	No	No
Programming	Possible depending on system	**Common**	Possible depending on system	No	No
Production volume	**High**	Medium–high	Low–medium	**High**	Medium
Initial investment	High	Medium–high	Medium	High	**High**
Fixture cost	**High**	Medium–high	Very low	Low	Low
Best stage	After PCBA assembly	End-of-line	Prototype/NPI/low volume	SMT/assembly	SMT/BGA/hidden joints